

Deep Research: Avatrada 57 Topic 001

Engineering Report: Dealer Gamma Exposure (GEX) Calculation and Analysis

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Subject: Deep-Dive on Dealer Gamma Exposure (GEX) Formula and Implementation

Executive Summary

This report provides a detailed engineering analysis of the Dealer Gamma Exposure (GEX) formula as specified in the source document. GEX is a critical metric used to estimate the hedging activity of options market makers (dealers), which in turn can influence market stability and volatility.

We will deconstruct the GEX formula, clarify the underlying assumptions about dealer positioning, and explain the crucial sign convention used in its calculation. An implementation strategy using Python and the Pandas library is provided, including a code snippet for calculating GEX across an entire options chain. Finally, the report conducts a critical analysis of the model, identifying its potential failure modes, limitations, and areas for optimization. The objective is to provide a comprehensive guide for accurately implementing and interpreting this financial metric.

1.0 Technical Deconstruction

The core of the analysis is the GEX formula and the assumptions that give it meaning.

1.1 The GEX Formula

The specified formula for calculating the dollar-denominated Gamma Exposure for a single options contract is:

$$\text{GEX}_{\{\$ \}} = \Gamma \times \text{OpenInterest} \times 100 \times S^2 \times 0.01$$

Let's break down each component:

- **Γ (Gamma):** This is a second-order option Greek. It measures the rate of change of an option's Delta (Δ) for a \$1 change in the underlying's spot price (S). Its unit is $\Delta / \$$. Gamma is always positive for long option positions (both calls and puts).
- **OpenInterest (OI):** This represents the total number of outstanding options contracts for a given strike and expiration. It is our proxy for the scale of market participation.
- **100:** The standard options contract multiplier. One equity option contract typically represents 100 shares of the underlying stock.
- **S (Spot Price):** The current market price of the underlying asset.
- **S^2 and 0.01:** These terms work together to convert the abstract Gamma value into a meaningful dollar amount representing the required hedging flow for a 1% move in the underlying. The derivation is as follows:
 1. **Total Gamma (in shares/\$):** $\Gamma \times \text{OI} \times 100$
 - This calculates the total change in the options position's share-equivalent delta for a \$1 move in the underlying.
 2. **Price Move for 1%:** $S \times 0.01$
 - This is the dollar magnitude of a 1% change in the spot price.
 3. **Shares to Hedge for a 1% Move:** $(\text{Total Gamma}) \times (\text{Price Move for 1\%})$
 - $(\Gamma \times \text{OI} \times 100) \times (S \times 0.01)$
 - This gives the total number of shares that must be bought or sold to re-establish a delta-neutral position after a 1% price change.

4. **Dollar Value of Hedge:** $(\text{Shares to Hedge}) \times S$

■ $[(\Gamma \times \text{OI} \times 100) \times (S \times 0.01)] \times S$

■ Rearranging the terms gives the final formula:

$\Gamma \times \text{OI} \times 100 \times S^2 \times 0.01$.

1.2 The Dealer Positioning Assumption

The entire GEX concept hinges on a crucial assumption: **Dealers (Market Makers) are the net counterparty to public/retail order flow.**

The rationale is: * **Public Behavior:** Retail and smaller institutional investors are typically net *buyers* of options. They buy calls for upside speculation and buy puts for downside protection (hedging). * **Dealer Role:** Market Makers provide liquidity by taking the other side of these trades. If the public is net long options, dealers are, by definition, net short options.

Therefore, the model assumes: * For Call options with significant Open Interest, the public is long, and **dealers are short**. * For Put options with significant Open Interest, the public is long, and **dealers are short**.

1.3 The Sign Convention and Hedging Dynamics

This is the most critical and nuanced part of the calculation. While dealers are assumed to be short both calls and puts (giving them negative Gamma from both), their hedging actions are directionally opposite.

- **Short Calls (Negative Gamma):** As the spot price S rises, the call's delta approaches +1. The dealer's short call delta becomes more negative (approaches -1). To remain delta-neutral, the dealer must **buy** the underlying asset. They chase the price up.
- **Short Puts (Negative Gamma):** As the spot price S falls, the put's delta approaches -1. The dealer's short put delta becomes more positive (approaches +1). To remain delta-neutral, the dealer must **sell** the underlying asset. They chase the price down.

Both scenarios *accelerate* the prevailing price trend, increasing volatility. To represent this in a single aggregate number, the SqueezeMetrics methodology introduces a sign convention:

The Gamma contribution from Put options is inverted (multiplied by -1).

- **Call GEX:** Calculated with a positive sign: $+ (\Gamma * OI * \dots)$
- **Put GEX:** Calculated with a negative sign: $- (\Gamma * OI * \dots)$

This convention aligns the hedging impact. After applying this convention, the resulting total GEX value can be interpreted:

- **Positive Total GEX:** Indicates a "long gamma" environment for dealers. This is counterintuitive to the assumption but arises when Put GEX dominates the calculation. In this regime, dealers sell into rallies and buy into dips, acting as a **stabilizing force** and suppressing volatility.
- **Negative Total GEX:** Indicates a "short gamma" environment. This is the typical state where Call GEX dominates. Dealers must buy into rallies and sell into dips, acting as an **accelerating force** and amplifying volatility.

Finally, since we are calculating **Dealer GEX**, and our initial calculation represents the public's exposure, we must invert the final sign. However, common practice is to label the aggregate number (with the put sign flipped) as "GEX" and interpret its sign directly as the market regime. We will follow this common practice in the implementation.

2.0 Implementation Strategy

We will use Python with the Pandas and NumPy libraries for an efficient, vectorized implementation.

Libraries: * `pandas`: For data manipulation and aggregation. * `numpy`: For efficient numerical operations, specifically for conditional sign flipping.

Data Schema: The input must be a Pandas DataFrame representing an options chain with at least these columns: * `strike`: The strike price of the option. * `type`: The option type, e.g., 'call' or 'put'. * `openInterest`: The open interest for the contract. * `gamma`: The gamma value for the contract.

The current `spot` price of the underlying is also required as a separate input.

Python/Pandas Implementation

The following function encapsulates the logic for calculating GEX and aggregating it by strike price.

```
import pandas as pd
import numpy as np

def calculate_gex(options_chain_df: pd.DataFrame, spot_price: float) ->
pd.DataFrame:
    """
    Calculates Gamma Exposure (GEX) for an entire options chain and aggregates
    by strike.

    Args:
        options_chain_df (pd.DataFrame): DataFrame with options data.
                                         Must contain ['type', 'gamma',
                                         'openInterest', 'strike'].
        spot_price (float): The current spot price of the underlying asset.

    Returns:
        pd.DataFrame: A DataFrame indexed by strike, showing the total GEX in
        dollars.
    """
    # --- 1. Define Constants ---
    CONTRACT_MULTIPLIER = 100.0
    PERCENT_MOVE_FACTOR = 0.01

    # --- 2. Handle potential missing data ---
    # Gamma or OI being NaN/None should result in zero exposure.
    df = options_chain_df.copy()
    df[['gamma', 'openInterest']] = df[['gamma', 'openInterest']].fillna(0)

    # --- 3. Calculate GEX for each option contract ---
    # This is the core formula: GEX = Gamma * OI * 100 * S^2 * 0.01
    gex = (
        df['gamma'] *
```

```

        df['openInterest'] *
        CONTRACT_MULTIPLIER *
        (spot_price ** 2) *
        PERCENT_MOVE_FACTOR
    )

    # --- 4. Apply the Sign Convention ---
    # Puts have their gamma contribution flipped to align hedging impact.
    # We create a sign multiplier: +1 for calls, -1 for puts.
    sign_multiplier = np.where(df['type'] == 'put', -1, 1)

    # This is the GEX from the public's perspective
    df['gex_public'] = gex * sign_multiplier

    # --- 5. Aggregate by Strike Price ---
    gex_by_strike = df.groupby('strike')['gex_public'].sum()

    # --- 6. Format Output ---
    # Dealer GEX is the inverse of Public GEX. We can analyze either.
    # For reporting, we'll show both total public and dealer GEX.
    total_public_gex = gex_by_strike.sum()
    total_dealer_gex = -total_public_gex

    print(f"Total GEX (Public Perspective): ${total_public_gex:,.0f}")
    print(f"Total GEX (Dealer Perspective): ${total_dealer_gex:,.0f}")
    print("----")
    print("Interpretation:")
    if total_dealer_gex > 0:
        print("Positive Dealer GEX Regime: Dealers may act as a stabilizing
force (sell rallies, buy dips).")
    else:
        print("Negative Dealer GEX Regime: Dealers may act as an accelerating
force (buy rallies, sell dips).")

    return gex_by_strike.to_frame('gex_dollars')

# --- Example Usage ---
# Create a sample options chain DataFrame
data = {

```

```

    'strike': [100, 105, 110, 115, 120, 100, 105, 110, 115, 120],
    'type': ['call', 'call', 'call', 'call', 'call', 'put', 'put', 'put',
'put', 'put'],
    'gamma': [0.05, 0.08, 0.10, 0.07, 0.04, 0.05, 0.08, 0.10, 0.07, 0.04],
    'openInterest': [5000, 8000, 12000, 7000, 3000, 6000, 9000, 10000, 6000,
2500]
}
sample_chain = pd.DataFrame(data)
underlying_spot_price = 112.50

# Run the calculation
gex_profile = calculate_gex(sample_chain, underlying_spot_price)

# Display the aggregated results
print("\nGamma Exposure Profile by Strike:")
print(gex_profile.apply(lambda x: x.map('{:,.0f}'.format)))

```

3.0 Critical Analysis

While GEX is a powerful conceptual tool, its implementation is subject to several limitations and potential failure modes.

3.1 Potential Failure Modes & Edge Cases

1. **The Core Assumption is a Generalization:** The model's foundation—that dealers are always the counterparty to a monolithic "public"—is its greatest weakness. In reality, the options market is a complex ecosystem of retail, hedge funds, and large institutions. A hedge fund selling covered calls has a different profile than a retail trader buying speculative calls. GEX is therefore a high-level proxy, not a precise measure of positioning.
2. **Data Quality and Latency:** The calculation is highly sensitive to the quality of its inputs.
 - **Stale Data:** Using stale `gamma` or `spot_price` values will lead to an inaccurate GEX snapshot. Gamma is dynamic and changes with price and volatility.

- **Inaccurate Open Interest:** OI data is typically updated once per day (End-of-Day). It does not reflect intraday position changes, which can be significant during volatile periods or on expiration days.
3. **Scope of Included Options:** The choice of which options to include is critical and can drastically alter the result.
- **Expiration Dates:** Including very long-dated options (LEAPs) can distort the GEX figure, as their near-term hedging impact is minimal. Most analyses focus on expiries within 30-90 days.
 - **Liquidity:** Illiquid, far out-of-the-money (OTM) options may have non-zero OI but negligible gamma and hedging impact. Including them adds noise. Filtering by a minimum OI or volume threshold is a common practice.

3.2 Optimizations and Best Practices

1. **Filtering:** Before calculation, the options chain should be filtered to a relevant scope. A good starting point is to include options expiring within the next 90 days and filter out contracts with zero open interest.
2. **Vectorization:** The provided Pandas/NumPy implementation is highly efficient as it avoids explicit loops in Python, leveraging vectorized operations for performance. This is the standard for this type of financial data processing.
3. **Dynamic Analysis:** GEX is not a static number. Its true power comes from observing how it changes over time and how the profile shifts as the underlying price moves. A robust system would calculate GEX periodically (e.g., every 5-15 minutes) to track the evolving hedging landscape.
4. **"Gamma Flip" Level:** A key level to monitor is the strike price where the aggregate GEX could "flip" from positive to negative (or vice-versa). This price level, often called the "Zero Gamma" level, can act as a pivot point in the market, where dealer hedging behavior is expected to change regimes. This can be found by calculating the cumulative sum of GEX across strikes.